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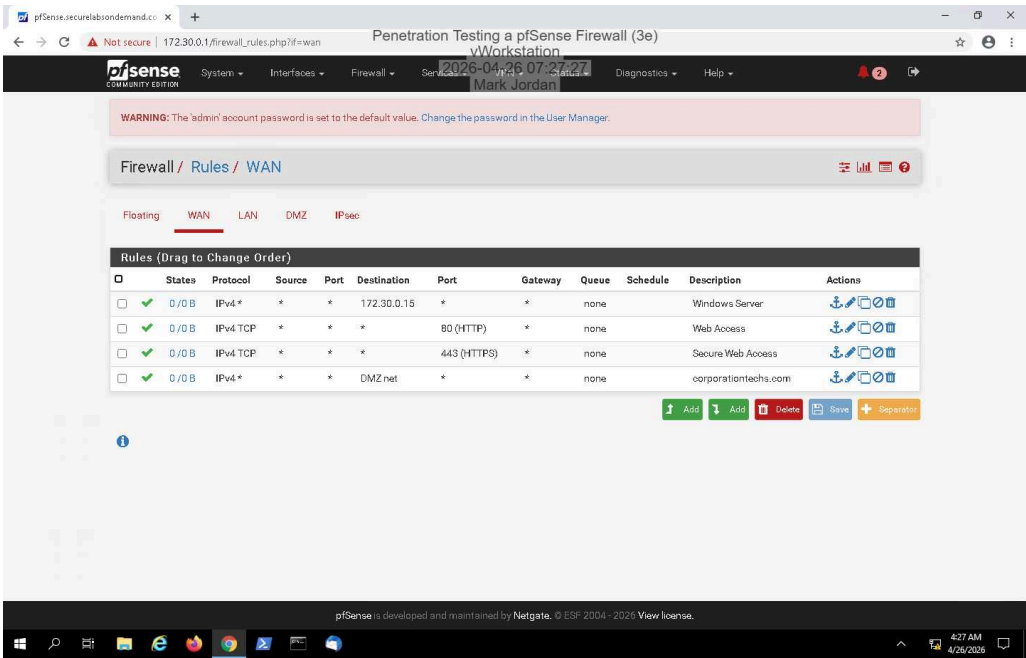
Time on Task:	Progress:
2 hours, 4 minutes	100%

Report Generated: Sunday, April 26, 2026 at 9:24 AM

Section 1: Hands-On Demonstration

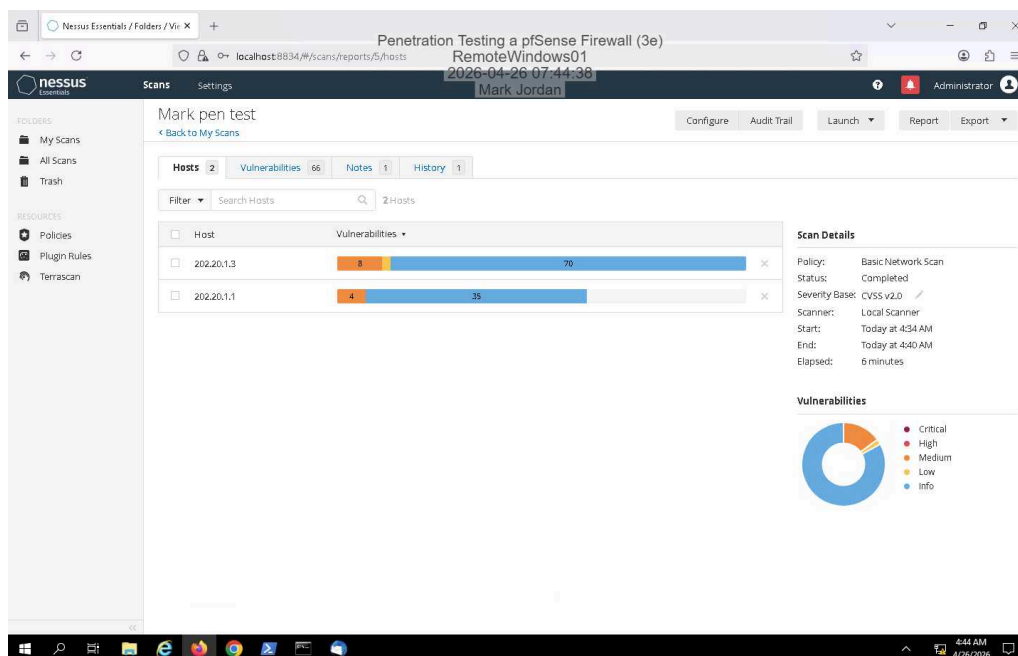
Part 1: Examine a pfSense Firewall Configuration

12. Make a screen capture showing the WAN rules table.

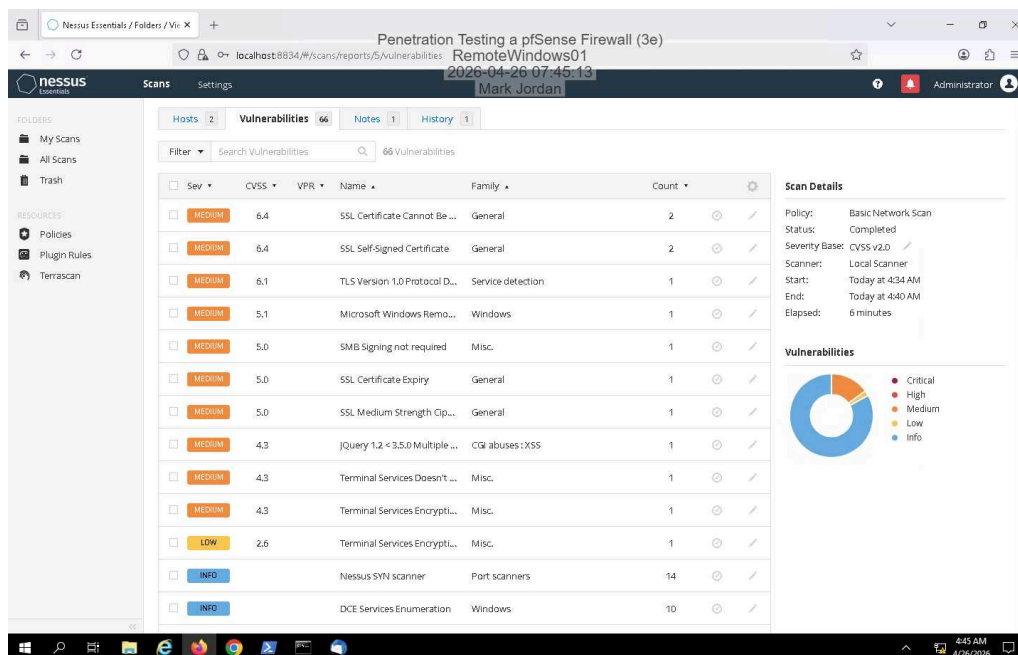


Part 2: Conduct a Penetration Test on the Network

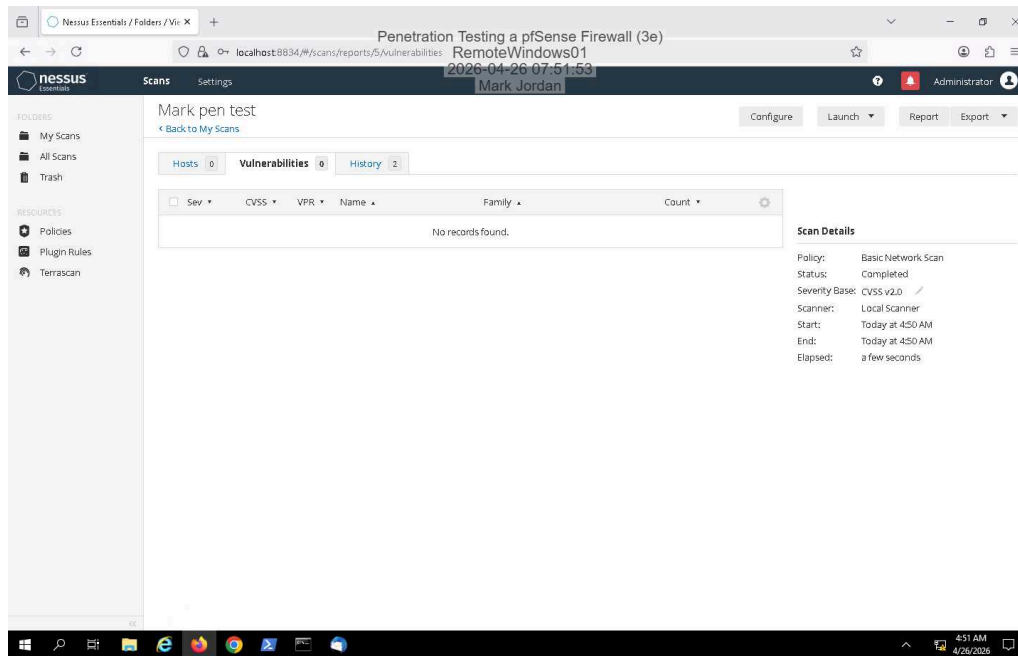
11. Make a screen capture showing the *yourname* pen test scan results.



13. Make a screen capture showing the list of vulnerabilities.



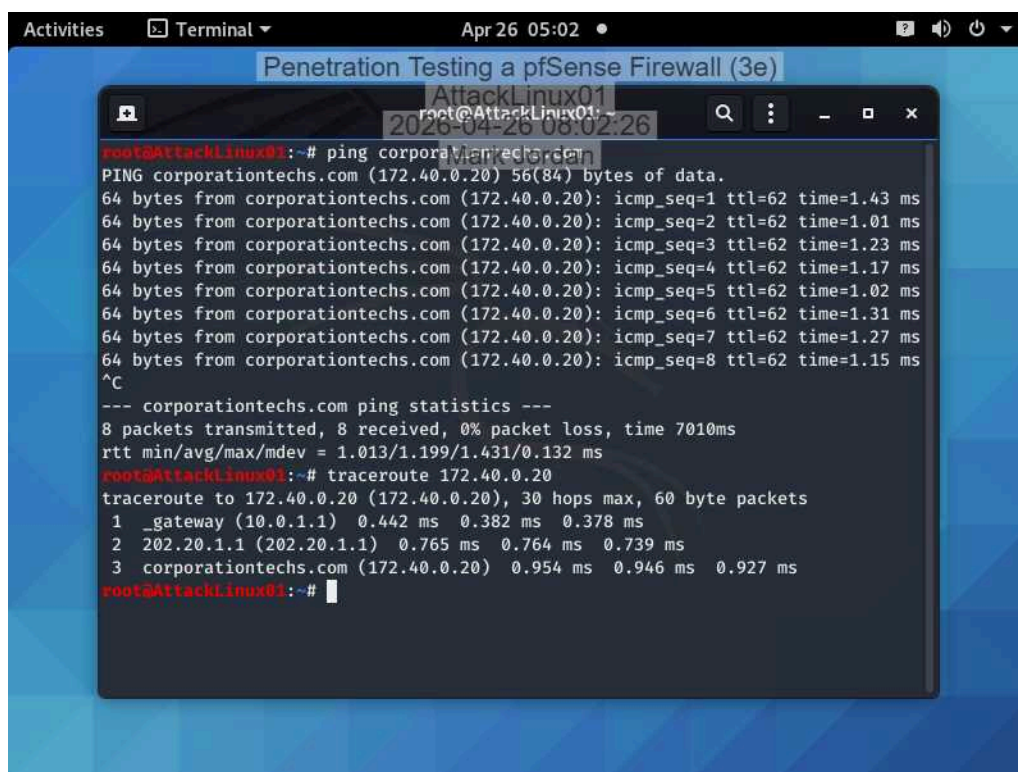
30. Make a screen capture showing the updated vulnerability report summary.



Section 2: Applied Learning

Part 1: Conduct a Port Scan on the Network

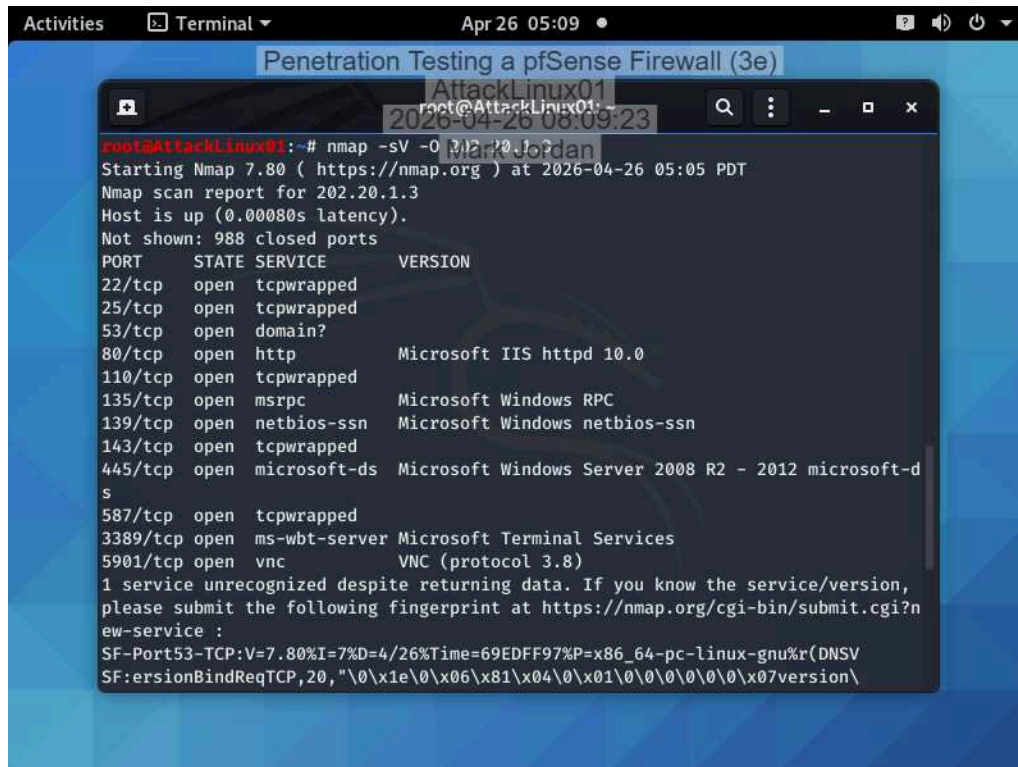
7. Make a screen capture showing the results of the traceroute command.



The screenshot shows a terminal window titled "Penetration Testing a pfSense Firewall (3e)" with a subtitle "AttackLinux01". The terminal output shows the results of a ping and traceroute command. The ping command was run against corporationtechs.com (172.40.0.20) and returned 8 successful responses with varying times. The traceroute command was run against 172.40.0.20 and showed a path of 3 hops: gateway (10.0.1.1), 202.20.1.1, and corporationtechs.com (172.40.0.20).

```
root@AttackLinux01:~# ping corporationtechs.com
PING corporationtechs.com (172.40.0.20) 56(84) bytes of data:
64 bytes from corporationtechs.com (172.40.0.20): icmp_seq=1 ttl=62 time=1.43 ms
64 bytes from corporationtechs.com (172.40.0.20): icmp_seq=2 ttl=62 time=1.01 ms
64 bytes from corporationtechs.com (172.40.0.20): icmp_seq=3 ttl=62 time=1.23 ms
64 bytes from corporationtechs.com (172.40.0.20): icmp_seq=4 ttl=62 time=1.17 ms
64 bytes from corporationtechs.com (172.40.0.20): icmp_seq=5 ttl=62 time=1.02 ms
64 bytes from corporationtechs.com (172.40.0.20): icmp_seq=6 ttl=62 time=1.31 ms
64 bytes from corporationtechs.com (172.40.0.20): icmp_seq=7 ttl=62 time=1.27 ms
64 bytes from corporationtechs.com (172.40.0.20): icmp_seq=8 ttl=62 time=1.15 ms
^C
--- corporationtechs.com ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7010ms
rtt min/avg/max/mdev = 1.013/1.199/1.431/0.132 ms
root@AttackLinux01:~# traceroute 172.40.0.20
traceroute to 172.40.0.20 (172.40.0.20), 30 hops max, 60 byte packets
 1 _gateway (10.0.1.1) 0.442 ms 0.382 ms 0.378 ms
 2 202.20.1.1 (202.20.1.1) 0.765 ms 0.764 ms 0.739 ms
 3 corporationtechs.com (172.40.0.20) 0.954 ms 0.946 ms 0.927 ms
root@AttackLinux01:~#
```

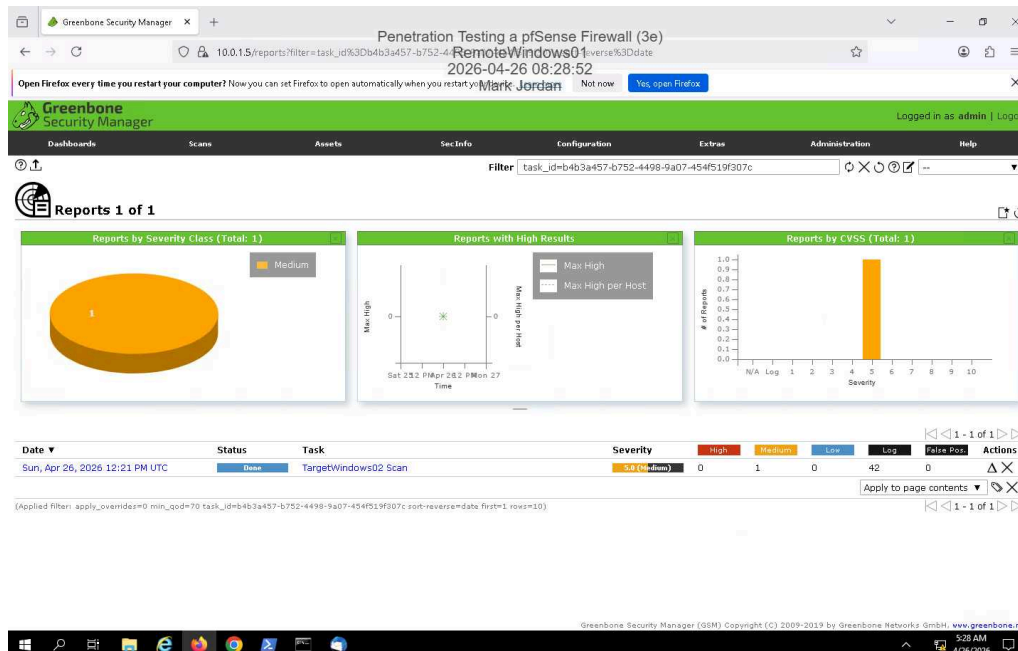
11. Make a screen capture showing the result of the nmap scan with OS detection activated.



```
root@AttackLinux01:~# nmap -sV -O 202.20.1.3
Starting Nmap 7.80 ( https://nmap.org ) at 2026-04-26 05:05 PDT
Nmap scan report for 202.20.1.3
Host is up (0.00080s latency).
Not shown: 988 closed ports
PORT      STATE SERVICE      VERSION
22/tcp    open  tcpwrapped
25/tcp    open  tcpwrapped
53/tcp    open  domain?
80/tcp    open  http         Microsoft IIS httpd 10.0
110/tcp   open  tcpwrapped
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
143/tcp   open  tcpwrapped
445/tcp   open  microsoft-ds Microsoft Windows Server 2008 R2 - 2012 microsoft-ds
587/tcp   open  tcpwrapped
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
5901/tcp  open  vnc          VNC (protocol 3.8)
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port53-TCP:V=7.80%I=7%D=4/26%Time=69EDFF97%P=x86_64-pc-linux-gnu%r(DNSV
SF:ersionBindReqTCP,20,"\\0\\x1e\\0\\x06\\x81\\x04\\0\\x01\\0\\0\\0\\0\\0\\0\\x07version\\
```

Part 2: Conduct a Vulnerability Scan on the Network

12. Make a screen capture showing the OpenVAS scan report.



14. Make a screen capture showing the detailed OpenVAS scan results.

The screenshot displays the Greenbone Security Manager (GSM) interface. The browser address bar shows the URL `10.0.1.5/report/591b7960-8330-4ad0-b62f-25328ad0b43f`. The page title is "Penetration Testing a pfSense Firewall (3e)". The user is logged in as "admin". The interface shows a navigation bar with tabs: Dashboards, Scans, Assets, Sec Info, Configuration, Extras, Administration, and Help. The main content area displays a report for "Sun, Apr 26, 2026 12:21 PM UTC". The report is titled "Report: Sun, Apr 26, 2026 12:21 PM UTC". The report ID is "591b7960-8330-4ad0-b62f-25328ad0b43f". The report was created on "Sun, Apr 26, 2026 12:21 PM UTC" and modified on "Sun, Apr 26, 2026 12:28 PM UTC". The report owner is "admin". The report shows a table of vulnerabilities with columns: Vulnerability, Severity, QoD, Host IP, Name, Location, and Created. The table lists several vulnerabilities, all with a severity of "0.0 (Low)" and a QoD of "80 %". The vulnerabilities include "CPE Inventory", "Hostname Determination Reporting", "Unknown OS and Service Banner Reporting", "Unknown OS and Service Banner Reporting", "Unknown OS and Service Banner Reporting", "Unknown OS and Service Banner Reporting", "Unknown OS and Service Banner Reporting", "CGI Scanning Consolidation", and "CGI Scanning Consolidation". The table also shows the host IP "202.20.1.3" and the location "general/tcp". The report is titled "Report: Sun, Apr 26, 2026 12:21 PM UTC".

Vulnerability	Severity	QoD	Host IP	Name	Location	Created
CPE Inventory	0.0 (Low)	80 %	202.20.1.3	general/CPE-T	general/tcp	Sun, Apr 26, 2026 12:27 PM UTC
Hostname Determination Reporting	0.0 (Low)	80 %	202.20.1.3	general/tcp	general/tcp	Sun, Apr 26, 2026 12:27 PM UTC
Unknown OS and Service Banner Reporting	0.0 (Low)	80 %	202.20.1.3	general/tcp	general/tcp	Sun, Apr 26, 2026 12:25 PM UTC
Unknown OS and Service Banner Reporting	0.0 (Low)	80 %	202.20.1.3	587/tcp	587/tcp	Sun, Apr 26, 2026 12:25 PM UTC
Unknown OS and Service Banner Reporting	0.0 (Low)	80 %	202.20.1.3	143/tcp	143/tcp	Sun, Apr 26, 2026 12:25 PM UTC
Unknown OS and Service Banner Reporting	0.0 (Low)	80 %	202.20.1.3	22/tcp	22/tcp	Sun, Apr 26, 2026 12:25 PM UTC
Unknown OS and Service Banner Reporting	0.0 (Low)	80 %	202.20.1.3	110/tcp	110/tcp	Sun, Apr 26, 2026 12:25 PM UTC
Unknown OS and Service Banner Reporting	0.0 (Low)	80 %	202.20.1.3	25/tcp	25/tcp	Sun, Apr 26, 2026 12:25 PM UTC
CGI Scanning Consolidation	0.0 (Low)	80 %	202.20.1.3	80/tcp	80/tcp	Sun, Apr 26, 2026 12:25 PM UTC
CGI Scanning Consolidation	0.0 (Low)	80 %	202.20.1.3	47001/tcp	47001/tcp	Sun, Apr 26, 2026 12:25 PM UTC

Section 3: Challenge and Analysis

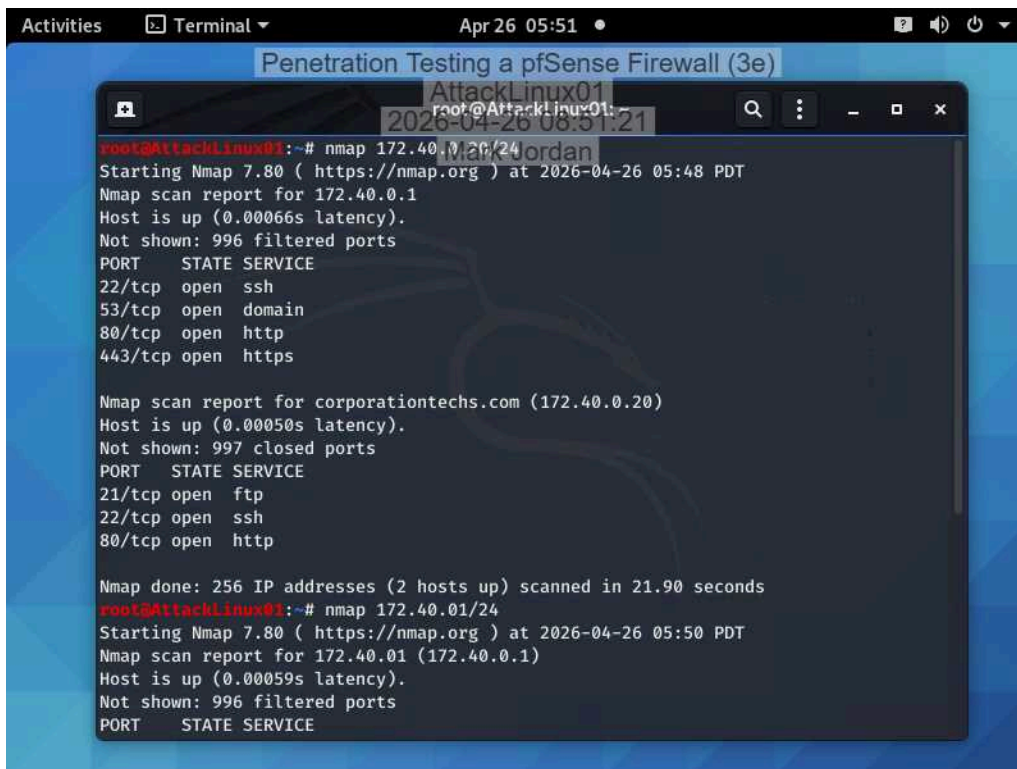
Part 1: Research DMZ Deployment Best Practices

Before beginning the technical portion of your penetration test, you decide to spend some time brushing up on best practices and common mistakes for DMZ deployments - both the network aspect and the servers located therein. Use the Internet to **research** DMZ deployments, then **identify** three best practices and one potential mistake or vulnerability.

The three best practices for DMZ deployment include: A dual stacked firewall architecture, applying strict deny by default rules, and applying real-time application layer inspection with WAFs or IPS's. Dual stacked firewall architecture means you are placing an external firewall between the internet and the DMZ and a second firewall placed between the DMZ and your LAN. With strict deny by default rules, this will help prevent any unwanted traffic that is not already whitelisted. Applying WAFs or IPS's will allow you to monitor and prevent malicious activity in real-time for an extra layer of protection. A common mistake with DMZ deployments is failing to account for the back door connections. Organization focus on hardening the internet to DMZ (front door) as oppose to the DMZ to the internal network (Back door)

Part 2: Conduct a Penetration Test on the DMZ

Make a screen capture showing the **open ports on the corporationtechs.com web server and the DMZ firewall interface.**

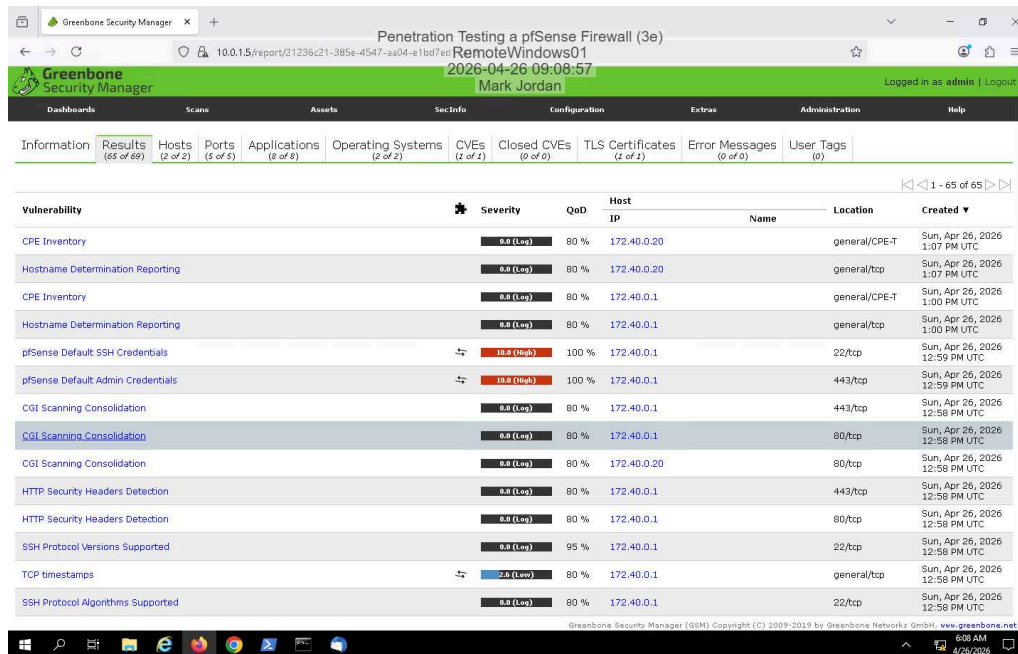


```
Activities Terminal Apr 26 05:51 Penetration Testing a pfSense Firewall (3e)
root@AttackLinux01:~# nmap 172.40.0.24/24
Starting Nmap 7.80 ( https://nmap.org ) at 2026-04-26 05:48 PDT
Nmap scan report for 172.40.0.1
Host is up (0.00066s latency).
Not shown: 996 filtered ports
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https

Nmap scan report for corporationtechs.com (172.40.0.20)
Host is up (0.00050s latency).
Not shown: 997 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http

Nmap done: 256 IP addresses (2 hosts up) scanned in 21.90 seconds
root@AttackLinux01:~# nmap 172.40.0.1/24
Starting Nmap 7.80 ( https://nmap.org ) at 2026-04-26 05:50 PDT
Nmap scan report for 172.40.0.1 (172.40.0.1)
Host is up (0.00059s latency).
Not shown: 996 filtered ports
PORT      STATE SERVICE
```

Make a screen capture showing the **vulnerability scan results**.



The screenshot displays the Greenbone Security Manager interface. The top navigation bar includes tabs for Dashboards, Scans, Assets, Sec Info, Configuration, Extras, Administration, and Help. The main content area shows a table of vulnerability scan results. The table has columns for Vulnerability, Severity, QoD, Host IP, Name, Location, and Created. The results are sorted by severity, with the highest severity (10.0) at the top. The table shows 65 results in total, with the first 10 displayed.

Vulnerability	Severity	QoD	Host IP	Name	Location	Created
CPE Inventory	0.0 (Log)	80 %	172.40.0.20		general/CPE-T	Sun, Apr 26, 2026 1:07 PM UTC
Hostname Determination Reporting	0.0 (Log)	80 %	172.40.0.20		general/tcp	Sun, Apr 26, 2026 1:07 PM UTC
CPE Inventory	0.0 (Log)	80 %	172.40.0.1		general/CPE-T	Sun, Apr 26, 2026 1:00 PM UTC
Hostname Determination Reporting	0.0 (Log)	80 %	172.40.0.1		general/tcp	Sun, Apr 26, 2026 1:00 PM UTC
pfSense Default: SSH Credentials	10.0 (High)	100 %	172.40.0.1		22/tcp	Sun, Apr 26, 2026 12:59 PM UTC
pfSense Default: Admin Credentials	10.0 (High)	100 %	172.40.0.1		443/tcp	Sun, Apr 26, 2026 12:59 PM UTC
CGI Scanning Consolidation	0.0 (Log)	80 %	172.40.0.1		443/tcp	Sun, Apr 26, 2026 12:58 PM UTC
CGI Scanning Consolidation	0.0 (Log)	80 %	172.40.0.1		80/tcp	Sun, Apr 26, 2026 12:58 PM UTC
CGI Scanning Consolidation	0.0 (Log)	80 %	172.40.0.20		80/tcp	Sun, Apr 26, 2026 12:58 PM UTC
HTTP Security Headers Detection	0.0 (Log)	80 %	172.40.0.1		443/tcp	Sun, Apr 26, 2026 12:58 PM UTC
HTTP Security Headers Detection	0.0 (Log)	80 %	172.40.0.1		80/tcp	Sun, Apr 26, 2026 12:58 PM UTC
SSH Protocol Versions Supported	0.0 (Log)	95 %	172.40.0.1		22/tcp	Sun, Apr 26, 2026 12:58 PM UTC
TCP timestamps	2.0 (Low)	80 %	172.40.0.1		general/tcp	Sun, Apr 26, 2026 12:58 PM UTC
SSH Protocol Algorithms Supported	0.0 (Log)	80 %	172.40.0.1		22/tcp	Sun, Apr 26, 2026 12:58 PM UTC

Part 3: Recommend Changes to the DMZ

Based on your research in Part 1 and your findings in Part 2, **prepare a brief summary** of recommended changes that Secure Labs on Demand should make to their DMZ deployment. Remember, your recommendations should apply to both the network configuration and the web server.

Based on the vulnerability scan. The top changes I recommend include: changing the pfSense default admin credentials as well as the pfSense default SSH credentials. These vulnerabilities have the highest severity of 10.0. Changing these default credentials will help slow down brute force attacks as they have to work harder to gain those credentials. Another vulnerability to fix would be the FTP Unencrypted Cleartext Login, an attacker can uncover login names and passwords by sniffing traffic to the FTP service. It is recommended to enable FTPS or enforce the connection via 'Auth TLS' command.